TECHNICAL & USER DOCUMENTATION

DWG / DXF Converter

QGIS Processing Plugin — Version 0.2.0

Author: Kelvin Moseti

Table of Contents

1. Introduction

DWG / DXF Converter is a QGIS plugin that adds bidirectional AutoCAD DWG ↔ DXF conversion to QGIS, implemented as four QGIS Processing algorithms. It wraps the free ODA File Converter command-line tool, so all the heavy lifting of actually decoding the proprietary DWG format is handled by Open Design Alliance's own software, this plugin's job is to drive that tool reliably, validate its results, and make the whole process a native part of the QGIS/Processing workflow.

Because it has no custom dialog of its own, every algorithm works identically whether you run it by hand from the Processing Toolbox, in bulk with QGIS's built-in batch runner, from a Processing model, from a PyQGIS script, or completely headlessly via the qgisprocess command-line tool.

2. Requirements & Installation

2.1 System Requirements

- QGIS 3.0 or later, including QGIS 4.x
- Windows, macOS, or Linux
- ODA File Converter (free, separate download — see below)
- Linux headless/server environments only: xvfb

2.2 Installing ODA File Converter

This plugin does not, and cannot, bundle a DWG reader or writer. DWG is a closed format, and no free or open-source tool may legally redistribute the libraries needed to read and write it. Instead, the plugin drives ODA's official, free converter.

1. Download the installer for your platform from: https://www.opendesign.com/guestfiles/oda_file_converter
2. Run the installer. No account, license key, or purchase is required.
3. On Windows, the default install path is typically under C:\Program Files\ODA\, the plugin searches recursively under Program Files, so a default install is normally found automatically.
4. On Linux servers or other headless environments, also install xvfb (for example: sudo apt install xvfb). ODA File Converter is a Qt GUI application internally and requires a display even for command-line batch conversion; the plugin automatically wraps calls with xvfb-run when no display is detected.

If the plugin cannot auto-detect the executable, its full path can be supplied directly via the “ODA File Converter executable” advanced parameter on any of the four algorithms.

2.3 Installing the Plugin

1. In QGIS: Plugins → Manage and Install Plugins → Install from ZIP.
2. Browse to the plugin's .zip file and click Install Plugin.
3. QGIS may warn that this is a third-party / unverified plugin, since it is not distributed through the official plugin repository — this is expected; confirm to proceed.
4. Open the Installed tab and make sure DWG / DXF Converter is checked/enabled.

Once enabled, four algorithms appear in the Processing Toolbox under the group “DWG / DXF Conversion”.

3. Algorithms Reference

3.1 Convert DWG file to DXF

Algorithm ID: dwg2dxf:convert_dwg_file

Converts a single DWG file to a single DXF file. Because it has exactly one file in and one file out, QGIS automatically offers “Run as Batch Process” for it (right-click the algorithm in the toolbox), making it usable for bulk work as well as single conversions.

3.2 Bulk convert DWG folder to DXF

Algorithm ID: dwg2dxf:convert_dwg_folder

Converts every DWG file in a folder to DXF in a single ODA File Converter invocation, faster than converting file-by-file for large batches. Skips files already converted and unchanged by default (see Section 6).

3.3 Convert DXF file to DWG

Algorithm ID: dwg2dxf:convert_dxf_to_dwg_file

The reverse direction: converts a single DXF file to AutoCAD DWG, useful for handing QGIS-produced or QGIS-edited work back to AutoCAD users. Since DWG files cannot be loaded as a QGIS layer, this algorithm does not offer the “add to project” option.

3.4 Bulk convert DXF folder to DWG

Algorithm ID: dwg2dxf:convert_dxf_to_dwg_folder

The reverse direction applied to a whole folder at once, with the same incremental skip-existing behaviour as the forward bulk algorithm.

3.5 Parameters Reference

Parameter	Applies to	Description
INPUT	both	Single file (file algorithms) or folder (bulk algorithms)
OUTPUT	both	Destination file or folder
OUT_VERSION	both	Target AutoCAD version for the result (e.g. ACAD2018); default ACAD2018
RECURSIVE	bulk only	Recurse into subfolders; default on
AUDIT	both	Ask ODA File Converter to audit/fix errors during conversion; default on
SKIP_EXISTING	bulk only	Incremental mode — skip files already converted and unchanged; default on
LOAD_TO_PROJECT	both	Add successful DXF results to the current project; only takes effect for DXF outputs
INPUT_FILTER	bulk only	Filename glob, supports ';'-separated patterns; default *.DWG or *.DXF
TIMEOUT_MINUTES	both	Advanced. Kill the process if it hasn't finished within this many minutes; 0 disables the timeout
ODA_PATH	both	Advanced. Overrides automatic detection of the ODA File Converter executable

4. Usage Guide

4.1 Interactive use (Processing Toolbox)

Open the Processing Toolbox (Processing menu → Toolbox, or Ctrl+Alt+T), expand DWG / DXF Conversion, and double-click any algorithm. QGIS generates the input dialog automatically from the algorithm's parameters, there is no custom UI to learn.

4.2 Batch mode over individual files

Right-click Convert DWG file to DXF (or its DXF-to-DWG counterpart) in the Processing Toolbox and choose Run as Batch Process. This lets you queue many individual file/output pairs and run them all in one go, a second bulk path alongside the folder-based algorithms.

4.3 Command line (qgis process)

```
# Single file, DWG -> DXF
qgis_process run dwg2dxf:convert_dwg_file \
  --INPUT=/data/plan.dwg \
  --OUTPUT=/data/plan.dxf

# Whole folder, DWG -> DXF, recursive, incremental by default
qgis_process run dwg2dxf:convert_dwg_folder \
  --INPUT=/data/incoming_dwgs \
  --OUTPUT=/data/converted_dxfs \
  --RECURSIVE=true

# Reverse direction, whole folder, DXF -> DWG
qgis_process run dwg2dxf:convert_dxf_to_dwg_folder \
  --INPUT=/data/dxfs_to_send_back \
  --OUTPUT=/data/dwgs_for_autocad
```

4.4 PyQGIS scripting

```
import processing

result = processing.run("dwg2dxf:convert_dwg_folder", {
    "INPUT": "/data/incoming_dwgs",
    "OUTPUT": "/data/converted_dxfs",
    "RECURSIVE": True,
    "SKIP_EXISTING": True,
    "TIMEOUT_MINUTES": 30,
})
print(result["OUTPUT"])
```

4.5 Processing models

Any of the four algorithms can be dragged into a Processing Model, chaining CAD conversion together with other Processing steps (for example, converting a folder of DWGs to DXF and then running a native QGIS algorithm over the results).

5. Reliability Features

All four algorithms share the same underlying implementation (two base classes: one for single-file, one for folder conversion), so every algorithm gets the same set of safeguards:

Timeout protection

ODA File Converter is occasionally known to stall on an unexpected blocking dialog (for example, a first-run prompt or an overwrite confirmation). Each algorithm enforces a configurable timeout (TIMEOUT_MINUTES); if the underlying process hasn't finished in that time, it is terminated and a clear error is raised, rather than the algorithm, and any pipeline calling it, hanging indefinitely.

Corrupt / empty output detection

A conversion that produces a 0-byte file is treated as a failure, not a silent success. This catches cases where ODA File Converter exits without actually writing a usable result.

Pre-flight validation (bulk algorithms)

Before invoking ODA File Converter, the bulk algorithms check that the input folder actually contains files matching the filename filter. If not, the algorithm fails immediately with a clear message rather than running ODA File Converter pointlessly and returning an empty, confusing result.

Per-file failure reporting (bulk algorithms)

If some files in a bulk run fail to convert, the Processing log names exactly which ones failed, rather than only reporting an aggregate count, making it straightforward to identify and re-check problem files.

6. Incremental ("Skip Already Converted") Mode

Bulk conversions default to skipping files that are already converted and unchanged, defined as: a same-named output file exists, is non-empty, and is at least as new as its source file. This has two practical effects:

- Re-running a bulk conversion over the same folder is fast, since only new or modified files are actually sent through ODA File Converter.
- It is safe to schedule a bulk conversion to run repeatedly (for example, nightly) against a folder that accumulates new files over time, without wasting time reconverting everything each run.

This behaviour is controlled by the SKIP_EXISTING parameter and can be turned off when a full, from-scratch conversion is required.

7. Loading Results into the QGIS Project

Both DWG-to-DXF algorithms offer a “Add resulting DXF to the current project” option, on by default. When enabled, every successfully converted DXF is loaded as a vector layer into the current project automatically once the algorithm finishes, no separate import step needed.

This option only applies to DXF outputs. DWG is not a format QGIS can read, so the two DXF-to-DWG algorithms do not offer it.

8. Troubleshooting

8.1 "ODA File Converter executable was not found"

The plugin could not locate ODA File Converter automatically. Locate ODAFileConverter.exe (Windows) or the equivalent binary manually (via File Explorer/Finder search, or 'which ODAFileConverter' on Linux/macOS), then paste the full path into the algorithm's ODA File Converter executable (advanced) parameter.

8.2 "This plugin is incompatible with this version of QGIS"

This means the plugin's metadata.txt does not declare compatibility with the installed QGIS version. This plugin declares qgisMinimumVersion=3.0 and qgisMaximumVersion=99.99, so it should install on any current QGIS 3.x or 4.x release; if this error appears, confirm you are installing the current version of the plugin zip rather than an older cached copy.

8.3 AttributeError referencing QProcess or Qgs... enum members

QGIS 4 uses PyQt6, which requires enum members to be accessed through their enum class (for example, QProcess.ExitStatus.NormalExit) rather than as a flat attribute (QProcess.NormalExit), which is how PyQt5 (QGIS 3) exposes them. This plugin resolves that automatically via a small compatibility helper (compat_enum) used throughout the codebase, so this class of error should not occur in the packaged plugin; if it does, it indicates an outdated copy of the plugin is installed.

8.4 Headless Linux: conversion hangs or fails immediately

ODA File Converter requires a display even in command-line/batch mode. On a headless Linux server, install xvfb (sudo apt install xvfb), the plugin automatically wraps the ODA File Converter call with xvfb-run when it detects no $DISPLAY is set. Without xvfb installed, the algorithm raises a clear error explaining this requirement.

8.5 Some files in a bulk conversion failed

Check the Processing log, failed files are listed by name. Common causes are corrupted, encrypted, or password-protected source files, or a DWG version newer than the installed ODA File Converter build supports. Try converting the specific failing file individually with the single-file algorithm for a more detailed error.

9. Limitations

- Conversion fidelity is entirely dependent on ODA File Converter; this plugin automates calling it, validates its results, and integrates them into QGIS, but does not alter how DWG/DXF entities are interpreted.
- Encrypted or password-protected DWG files will fail to convert.
- There is no fine-grained, per-entity progress bar during conversion, since ODA File Converter's command-line interface does not expose one; the Processing log shows raw tool output plus summary counts instead.

10. Appendix: Supported Output Versions

The OUT_VERSION parameter accepts any of the following AutoCAD version targets, corresponding to ODA File Converter's supported output versions:

- ACAD9, ACAD10, ACAD12, ACAD13, ACAD14
- ACAD2000, ACAD2004, ACAD2007, ACAD2010, ACAD2013, ACAD2018 (default)

— End of document —